Benchmark Items as Measurement Instruments: Harmonized DIF Diagnostics for BBH Model Regimes

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Abstract

This project studies BBH benchmark items as measurement instruments rather than only leaderboard components. Models are treated as respondents, benchmark questions as items, and the main question is whether item functioning is invariant across model regimes. Using a harmonized design on Open LLM Leaderboard BBH data, I compare a family-label baseline, a strict same-wrong MADNESS-inspired grouping, and an ability-neutral response-derived grouping (EP-1b). The evidence is cautious. On a common held-out 1,727-item DIF-test split, the family baseline shows broad Llama–Qwen noninvariance, EP-1b shows a similar but somewhat weaker held-out DIF surface while preserving response-derived grouping, and strict MADNESS remains diagnostically problematic because it is strongly ability-skewed despite a large DIF signal. Held-out consequence analyses show nonzero item-pool effects for all three analyses, but these results do not establish that global leaderboards are invalid. In this BBH testbed, benchmark item functioning can differ across regimes; the psychometric contribution is a diagnostic lens on benchmark validity rather than a universal claim about all benchmarks.

1 Introduction and motivation

LLM benchmarks are typically summarized as aggregate scores, but the underlying data are item responses [Suzgun et al., 2023, Liang et al., 2022]. This suggests a measurement framing: models are respondents, benchmark questions are items, and benchmark validity depends partly on whether items function comparably across the model regimes being compared. If an item is easier for one regime than another after conditioning on overall performance, it exhibits differential item functioning (DIF) [Holland and Thayer, 1988]. Under that framing, leaderboards are not assumed invalid by default; instead, they become instruments whose invariance should be checked.

This report asks whether BBH item functioning is invariant across model regimes in a harmonized Open LLM Leaderboard testbed. The key claim is deliberately cautious: benchmark item functioning can differ across regimes in this BBH setting, but that does not by itself prove that full leaderboards are globally invalid. The stronger contribution is methodological. Earlier project iterations used partly mismatched discovery and testing schemes across the family-label, strict MADNESS, and EP-1b analyses. The current report aligns them to a common model split and a common item split so that the three analyses can be compared on the same DIF-test item set.

2 Harmonized design and methods

The final merged BBH response matrix contains 4,409 models and 5,760 items. Models are split once into 3,179 discovery models and 1,230 held-out evaluation models using a fixed-seed, proxy

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Table 1: Harmonized model/item split diagnostics.

Quantity	Count
Discovery models	3179
Evaluation models	1230
Group-discovery items	4033
DIF-test items	1727
Family baseline discovery: Llama	1097
Family baseline discovery: Qwen	820
Family baseline evaluation: Llama	355
Family baseline evaluation: Qwen	335
Strict MADNESS decoded group-discovery items retained	3026

cluster-aware 70/30 split. Items are split once into 4,033 group-discovery items and 1,727 DIF-test items using a fixed-seed 70/30 domain-stratified split.

The harmonized design has two principles. First, all three analyses use the same held-out DIF-test item set for MH-DIF. Second, response-derived groups are learned only from discovery models on group-discovery items. Thus the 1,727 DIF-test items are not used in strict MADNESS or EP-1b grouping, tuning, centroid construction, or evaluation-model assignment.

Family-label DIF baseline. The family baseline uses external Llama/Qwen labels and no group-learning step. MH-DIF is run on discovery Llama/Qwen models using only the DIF-test items. This yields 1,097 Llama and 820 Qwen discovery models, plus 355 Llama and 335 Qwen held-out evaluation models. Anchor20 is implemented as: initial total-score MH pass, select the lowest-$|\Delta_{MH}|$ 20% of items as anchors, re-match using anchor-only scores, then rerun MH with BH-FDR [Mantel and Haenszel, 1959, Holland and Thayer, 1988, Benjamini and Hochberg, 1995]. However, the implementation is still a simplified Hansol-style replication: cluster-robust standard errors and Rao–Scott / design-effect corrections are not implemented.

Strict MADNESS. The pre-analysis response-derived starting point was strict same-wrong clustering inspired by a Madness-of-Crowds style idea: if two models repeatedly choose the same wrong option, they may occupy a shared response regime. In the harmonized version, groups are learned only from discovery models on group-discovery items with decoded options available. Of the 4,033 group-discovery items, 3,026 are retained for strict MADNESS after decoded-option filtering. The rule is exact: items must have decoded options with option count at least 2 and at least two retained models with valid decoded cells; models are retained only if they have at least 2,420 valid decoded group-discovery cells, i.e., $\max(50, 0.8 \times 3026)$. This keeps 4,399 models and drops 10 for low decoded coverage.

Ward-style hierarchical clustering was applied to a dissimilarity derived from same-wrong z-similarity. Held-out evaluation models are then assigned using only group-discovery items and a frozen rule: each evaluation model is mapped to the group with higher mean same-wrong z-similarity to discovery members.

EP-1b. EP-1b is the refined response-derived analysis. It preserves endogenous grouping while explicitly reducing trivial ability separation. Features are residualized domain-response shapes computed from discovery models on group-discovery items only. Residualization removes a cubic function of overall score, the discovery models are clustered by k-means, and the selected solution is then frozen. Held-out evaluation models are assigned using only group-discovery items by applying the same residualization/standardization transform and assigning each model to the nearest frozen discovery centroid.

Consequentiality. Consequences are evaluated only on held-out evaluation models. For each analysis, DIF categories learned on discovery models define selected item pools on the common DIF-test item set. The consequence statistic is the focal-minus-reference held-out accuracy gap on those item pools. The sign convention is Qwen$-$Llama for the family baseline, G1$-$G0 for strict MADNESS, and EPn1$-$EPn2 for EP-1b.

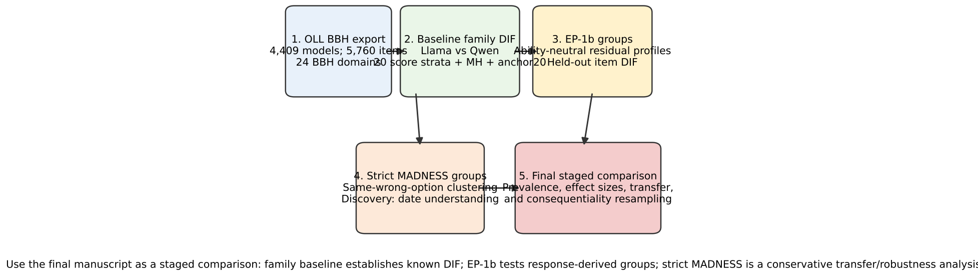

Figure 1: Overall analysis logic. The harmonized report keeps the same high-level narrative but forces all analyses onto the same model split and the same DIF-test item split.

Table 2: Harmonized MH-DIF results on the common 1,727-item DIF-test split.

Analysis	Items	A	B	C	Median $\lvert\Delta_{MH}\rvert_C$	Usable strata
Family baseline	1727	617 (35.7%)	258 (14.9%)	852 (49.3%)	2.47	20/20
Strict MADNESS	1727	461 (26.7%)	185 (10.7%)	1081 (62.6%)	3.28	11/20
EP-1b	1727	674 (39.0%)	307 (17.8%)	746 (43.2%)	2.27	20/20

3 Results

Family-label DIF baseline. The family baseline remains the clean sanity-check result. On the common DIF-test set, it finds broad noninvariance. Among 1,727 DIF-test items, 617 (35.7%) are ETS A, 258 (14.9%) are ETS B, and 852 (49.3%) are ETS C, with median $|\Delta_{MH}| = 2.47$ among Category-C items. This confirms that the BBH parsing and MH/ETS pipeline still recover strong family-level noninvariance under the stricter harmonized split.

Strict MADNESS. The harmonized strict MADNESS run is informative but still not the clean main result. It does produce a strong DIF surface. On the same 1,727 DIF-test items, it yields 461 A items (26.7%), 185 B items (10.7%), and 1,081 C items (62.6%), with median $|\Delta_{MH}| = 3.28$ among Category-C items. However, this signal is hard to interpret as regime-specific item functioning rather than broad ability separation.

The main diagnostic problem is matching. Only 11 of 20 anchor strata are usable. The low-score tail is almost entirely G0 and the high-score tail almost entirely G1, so many bins fail the minimum-per-group requirement. This is why strict MADNESS remains diagnostically awkward even after harmonization: the DIF percentages are large, but the grouping still behaves partly like an ability split.

EP-1b. EP-1b remains the strongest response-derived analysis because it stays fully within the harmonized item split while avoiding the extreme stratum collapse seen in strict MADNESS. On the discovery split, EPn1 has 1,621 models and EPn2 has 1,558; on the held-out evaluation split, EPn1 has 645 and EPn2 has 585. Unlike strict MADNESS, EP-1b assigns all models. Its held-out DIF profile on the common 1,727-item test set is 674 A items (39.0%), 307 B items (17.8%), and 746 C items (43.2%), with median $|\Delta_{MH}| = 2.27$ among Category-C items and all 20 strata usable. This is somewhat weaker than the family baseline in Category-C prevalence, but it is cleaner as a response-derived analysis because it uses a frozen discovery-only centroid assignment and avoids the severe matching collapse of strict MADNESS.

Consequentiality. The consequence analysis shows nonzero selected-item-pool effects without supporting a claim of global leaderboard invalidity. For the family baseline, the full-pool held-out Qwen$-$Llama gap on DIF-test items is 2.55 percentage points; selected B/C pools move that gap to +18.70 or -12.33 points depending on direction. For EP-1b, the full-pool EPn1$-$EPn2 gap is only 1.33 points, but selected B/C pools move it to +13.63 or -11.38 points. Strict MADNESS also shows large pool-specific shifts, but because the full-pool G1$-$G0 gap is already 18.29 points

Table 3: Strict MADNESS anchor20 stratum diagnostics. Only 11 of 20 bins are usable because the low and high score tails are dominated by one group.

Score bin	G0	G1	Usable
0	166	0	No
1	163	0	No
2	167	0	No
3	147	0	No
4	155	1	No
5	147	7	No
6	120	41	Yes
7	133	40	Yes
8	120	60	Yes
9	97	60	Yes
10	73	69	Yes
11	54	102	Yes
12	67	114	Yes
13	73	64	Yes
14	91	88	Yes
15	33	105	Yes
16	11	146	Yes
17	0	176	No
18	1	133	No
19	6	144	No

Table 4: Response-derived group sizes in the harmonized split.

Split	G0	G1	EPn1	EPn2	Unassigned
ep1b_discovery	0	0	1621	1558	0
ep1b_evaluation	0	0	645	585	0
strict_madness_discovery	1824	1350	0	0	5
strict_madness_evaluation	310	915	0	0	5

and even G0-favoring pools remain positive, those consequences are confounded with broad group-performance differences.

4 Discussion

The harmonized rerun clarifies the project story. First, the family baseline remains strong under a stricter split, so the underlying DIF machinery is not a fragile artifact of earlier staging choices. Second, strict MADNESS still contains useful response-derived structure, but the harmonized diagnostics show why it is not the best main analysis: many score bins are unusable, the groups are performance-skewed, and the held-out consequence pattern does not cleanly separate item-level noninvariance from broad group ability. Third, EP-1b remains the best response-derived analysis because it uses only discovery models and group-discovery items for learning, applies a frozen evaluation assignment rule, preserves all 20 strata, and still shows a substantial held-out DIF surface on the common test set.

This staged pattern supports a cautious conclusion. In this BBH testbed, item functioning is not invariant across model regimes. Response-derived grouping can recover that fact even without external family labels, but only some grouping strategies are psychometrically interpretable. Strict same-wrong clustering is attractive because it is purely endogenous, yet in this run it is too entangled with broad performance differences. EP-1b is less dramatic but more credible as an ability-neutral response-derived analysis.

The limitations remain important. The benchmark is BBH-only, not a representative sample of all LLM evaluations. Strict MADNESS depends on decoded-option availability and drops items lacking reliable option parsing. The family baseline is only a simplified Hansol-style MH/ETS replication because cluster-robust standard errors and Rao–Scott corrections are not implemented. Group definitions are method-sensitive, as the contrast between strict MADNESS and EP-1b shows.

Table 5: Held-out evaluation consequence analysis. Gap signs are Qwen−Llama, G1−G0, and EPn1−EPn2.

Analysis	Pool	Pool items	Eval models	Full-pool gap pp	50-item draw mean (SD) pp
family_llama_qwen	All DIF-test items	1727	690	2.55	2.57 (2.41)
family_llama_qwen	Invariant A	617	690	1.12	1.11 (1.25)
family_llama_qwen	B/C favor Qwen	561	690	18.70	18.73 (1.85)
family_llama_qwen	B/C favor Llama	549	690	-12.33	-12.39 (1.51)
strict_madness_ward	All DIF-test items	1727	1225	18.29	18.35 (3.25)
strict_madness_ward	Invariant A	461	1225	17.84	18.03 (2.95)
strict_madness_ward	B/C favor G1	673	1225	28.39	28.42 (2.85)
strict_madness_ward	B/C favor G0	593	1225	7.16	7.19 (2.94)
ep1b	All DIF-test items	1727	1230	1.33	1.29 (1.68)
ep1b	Invariant A	674	1230	0.89	0.86 (0.69)
ep1b	B/C favor EPn1	547	1230	13.63	13.66 (1.07)
ep1b	B/C favor EPn2	506	1230	-11.38	-11.42 (0.95)

Finally, consequentiality is evaluated here at the item-pool level, not as a full global rank-stability analysis.

5 Conclusion

This harmonized rerun strengthens the paper's core claim while narrowing its scope. BBH benchmark items can function differently across model regimes, and this remains visible when family-label, strict MADNESS, and EP-1b analyses are forced onto the same model split and the same held-out DIF-test item set. The family baseline is strong, strict MADNESS is informative but too ability-skewed to lead the story, and EP-1b is the best main response-derived analysis under the current evidence. The safest interpretation is that measurement noninvariance is a useful diagnostic lens for LLM benchmark validity, not a blanket argument that all leaderboards are invalid.

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